

Hangulo: Demonstrating Workflow-Embedded AI Support for Korean Lettering Implementation

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Abstract

The design process is commonly understood as an iterative cycle of ideation, implementation, and evaluation. During implementation, designers refine and develop their ideas into meaningful outcomes through ongoing micro-level reflection. Unlike tools that automate this process in a single step, we present Hangulo, a Korean lettering tool featuring workflow-embedded agents, demonstrating how AI can augment designers' in-progress edits within existing implementation workflows. Leveraging Hangul's compositional structure, Hangulo propagates style edits across structurally related letters and translates rough sketches into vector-level suggestions, replacing tedious vector control steps while preserving opportunities for micro-level reflection throughout implementation.

CCS Concepts

• **Applied computing** → *Media arts*; • **Human-centered computing** → *Interactive systems and tools*; • **Computing methodologies** → *Intelligent agents*.

Keywords

lettering, typography, creative workflow, AI agents, implementation support, idea augmentation

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1 Background

Design typically unfolds as an iterative cycle of ideation, implementation, and evaluation. Prior work has extensively explored ideation [1, 3, 4, 13] and evaluation [2, 9], leaving implementation, where designers turn raw ideas into outcomes, comparatively underexplored. Many design agents today generate outcomes in a single pass from a text prompt, a pattern also seen in letter design, where current AI approaches focus mainly on image-level stylization [11, 12, 14, 16, 17] or outcome steering [10, 15], with designers adjusting results rather than constructing letterforms

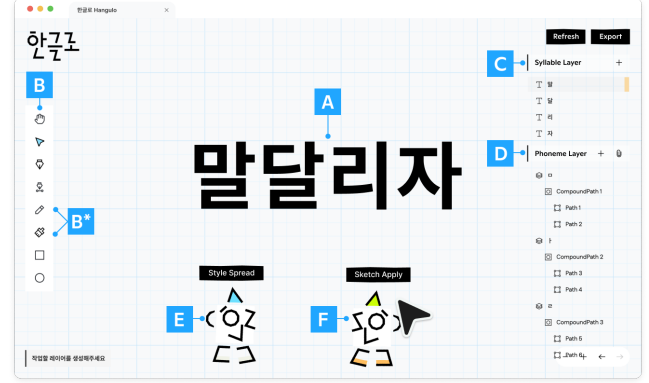


Figure 1: Interface of Hangulo

as structured entities. Yet implementation rarely happens in one step. It unfolds through many micro-actions, each involving its own reflection-in-action [7], and collapsing it into a single upfront specification leaves little room for this reflection. In this work, we explore how AI's generative capabilities can be harnessed at the level of micro-actions, augmenting each step as it happens so that designers retain moment-to-moment reflection while still benefiting from automation.

We ground our exploration in Korean lettering, a domain with strong structural constraints and rich expressive possibilities. Hangul's 19 consonants and 21 vowels all derive from 5 basic consonant forms and 3 basic vowel forms, assembling into 6 compositional block structures where a consonant's shape shifts with its block composition [5]. This shared, decomposable structure breaks lettering work into smaller, organically related units, a single stroke, a phoneme, a syllable, making it a fitting domain to demonstrate implementation support at this level. To identify concrete needs, we reviewed lettering communities and tutorials, surveyed practitioners (N=8), and validated findings with a professional designer (self-reported proficiency 4/5). Designers typically sketch, vectorize promising ideas, and iteratively refine through variations, a non-linear process where they may return to sketching mid-refinement. We identified two pain points: extending a stylistic idea to structurally different phonemes while balancing consistency and variation (e.g., a style for 'ㄴ' extends easily to 'ㄷ', sharing its triangular form, but not to 'ㄹ'), and cumbersome vectorized-letterform editing that discourages exploring small variations. These shaped two design goals:



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reduce tedious vector manipulation, and preserve micro-level reflection by augmenting emerging ideas rather than replacing designers' decision-making.

2 System

Hangulo is a web-based vector editing application built with Paper.js¹, featuring six standard vector operations and two sketching operations (Figure 1B-B*). Once a designer inputs the target word, the system constructs a layer for each syllable and phoneme character, starting from an empty canvas or an imported base font or SVG file. The two AI agent features, invoked from the characters at the bottom of the canvas (Figure 1E-F), use Claude Haiku for interpretation and planning and Claude Sonnet for execution.

2.1 Style Spread

Style Spread enables designers to maintain stylistic consistency across Hangul letters by propagating style edits to structurally related characters, rather than treating each letter in isolation, following the pipeline shown in Figure 2. (A) Once a designer modifies the style of a specific consonant or vowel (here, 'ㄴ'), they trigger this feature (Figure 1E), and the pre- and post-edit path data enter the agent pipeline as SVG text. (B) The edit interpretation module infers the designer's intention, analyzing the geometric and semantic transformation (e.g., "...removes the rigid stem-dot identity of 'ㄴ' and replaces it with more fluid, curved decorative elements..."). (D) The form similarity computation module then computes a similarity distance between the edited jamo ('ㄴ') and each candidate jamo ('ㅁ', 'ㄷ', 'ㄹ', 'ㄴ') using the (C) annotated Hangul jamo dataset (see Appendix), which categorizes all consonants and vowels into one of nine form groups (e.g., square-group, circular-group, vertical-line-group) and encodes morphological similarities across 18 form-level features (e.g., double-corner, diagonal-line, double-tick, horizontal-line), letting the agent reason about shared geometric characteristics across letters. (E) Designers then review and select a propagation range based on these similarity scores. (F) The planning module drafts how the style change should be applied to each jamo (e.g., "transform the lower straight stroke of 'ㄷ' into a curved ribbon-like stroke, extending the curved motion introduced in 'ㄴ'"), allowing designers to review the reasoning before it is applied. (G) The execution module then carries out the plan, and (H) designers review the generated outputs, selectively importing results into the canvas or requesting alternatives.

2.2 Sketch Apply

Sketch Apply enables designers to communicate raw stylistic intentions through informal sketches that guide the agent in generating deformed vectors. Our formative interview identified two recurring sketch types in letter design: (1) skeleton sketches, expressing how a stroke's axis should change (e.g., bend, wave, tilt, weave), and (2) outline sketches, expressing how a stroke's form should be shaped (e.g., adding decoration or an expressive cap). As shown in Figure 3, designers draw these sketches directly over a phoneme letter using two tools (Figure 1B*): *Marker* for skeleton sketches and *Pencil* for outline sketches. (A-B) The sketch path and the target jamo's path data are passed to the agent pipeline as SVG text. (C) The sketch

interpretation module reads the sketch to infer the designer's intended transformation. (D) The execution module then generates two candidate vectors per interpretation, reflecting distinct readings of the sketch. (E) Designers review the candidates, select one, and import it into the canvas.

3 Conclusion

This work presented Hangulo, a design demonstration exploring how AI can support the implementation stage of creative practice by augmenting designers' evolving inspirations. Rather than generating final outcomes through textual instructions, Hangulo integrates AI into ongoing Korean lettering workflows in a human-in-the-loop manner, responding to intermediate artifacts such as edit histories and sketches. This reduces tedious vector manipulation while preserving continuous engagement in micro-level reflection throughout implementation.

This perspective opens several directions for future work. One direction is to examine how such micro-level implementation agents affect designers' sense of agency, authorship, and authenticity, and how designers come to use and perceive these agents within their own workflows over time. Second, our approach is itself part of a broader effort to track and interpret design actions, already underway across several domains [6, 8]. An open question in this line of work is what support designers need beyond an agent's own inference, and what forms of interaction can help resolve the mismatch between an agent's interpretation and a designer's actual intent.

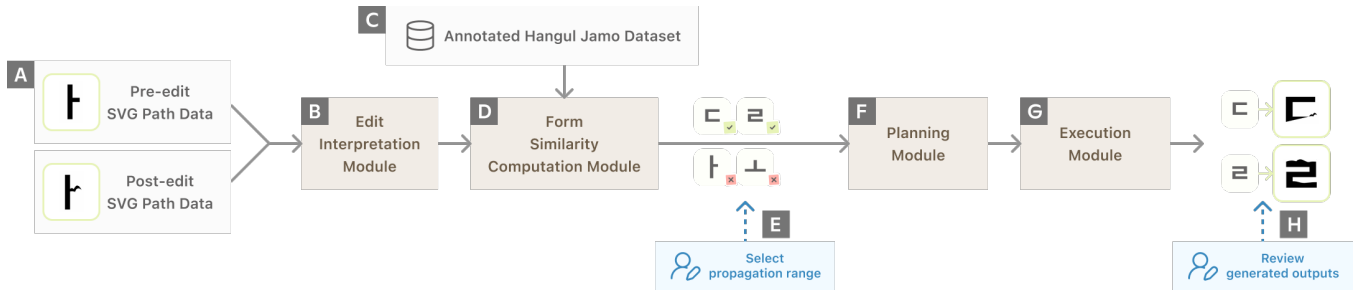
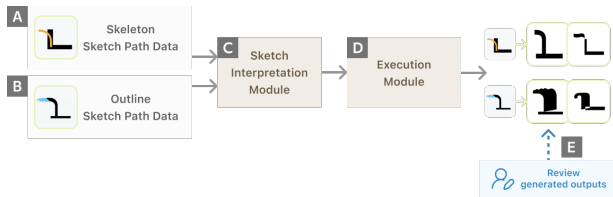
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¹<http://paperjs.org/>


 Figure 2: Agent pipeline of *Style Spread*

 Figure 3: Agent pipeline of *Sketch Apply*

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